

A Project

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Learning outcomes

Numerical system in Haskell:

- binary notation for `Positive`
- is an instance of `Num`
- requires to be instance of `Eq`
- it is also an instance of `Ord`

Numerical system in Binary notation

How to define a numerical system.

```
type Num :: * -> Constraint
class Num a where
    (+) :: a -> a -> a
    (-) :: a -> a -> a
    (*) :: a -> a -> a
    negate :: a -> a
    abs :: a -> a
    signum :: a -> a
    fromInteger :: Integer -> a
```

Implementing Positive

- Numbers.Positive.Structure.hs
- Numbers.Positive.OperationsUnary.hs
- Numbers.Positive.VsInteger.hs

Observations

- Extending `Positive` with a single element yields $\mathbb{N}$ in binary notation
- `Positive` can be instance of `Eq` and of `Ord`
- `succ`, `pred :: Positive -> Positive -> Positive` have the obvious definition